CHAPTER-3 MATRICES AND DETERMINANT

Multiple Choice Questions

1. If A and B are matrices then $(AB)^t =$

(A) $B^t A^t$ (B) $A^{-1}B^{-1}$ (C) AB (D) none of these

2. A square matrix A is skew hermitian if $(\overline{A})^t =$

(A) A (B) $\overline{A}$ (C) A^t (D) – A

3. The inverse of a square matrix A exists if A is

(A) Singular (B) symmetric (C) non – symmetric (D) non – singular

4. The matrix $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 6 \end{vmatrix}$ is called.

(A) upper triangular (B)lower triangular (C) diagonal (D) scalar

5. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then adj A equals

(A) $\begin{bmatrix} d & b \\ c & d \end{bmatrix}$ (B) $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (C) $\begin{bmatrix} d & b \\ c & a \end{bmatrix}$ (D) $\begin{bmatrix} d & -b \\ c & a \end{bmatrix}$

6. If $A = \begin{vmatrix} 1 & -2 & 3 \\ -2 & 3 & 1 \\ 4 & -3 & 2 \end{vmatrix}$ then A_{33} equals

(A) – 1 (B) 1 (C) 7 (D) – 7

7. A square matrix is symmetric if A^t equals

(A) A (B) – A (C) A^2 (D) $\frac{A}{2}$

8. Let $A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$ then $|A|$ equals

(A) 4 (B) 3 (C) 5 (D) 7

9. $(AB)^{-1} =$

(A) AB (B)BA (C) $A^{-1}B^{-2}$ (D) $B^{-1}A^{-1}$

10. A homogenous system of linear equation has

(A) finite solutions (B) unique solutions (C) infinite (D)two solutions

11. If $A = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$ then cofactor of 6 is

(A) 1 (B) – 1 (C) – 6 (D) 3

12. If $|A|$ is the determinant of a square matrix A then $|A|$ is

(A) always positive (B) modulus of A (C) always – ve (D) may be + ve or – ve

13. $[0]$ is a

(A) square matrix (B) unit matrix (C) rectangular matrix (D) scalar matrix

14. For a square matrix A, $|A|$ equals

(A) A (B) $|A^t|$ (C) $-|A^t|$ (D) – A^t

15. **If** $A=\begin{bmatrix} a & d \\ b & c \\ c & f \end{bmatrix}$ **then order of A^t is**

(A) 3 x 2 (B)2 x 3 (C) 2 x 2 (D) 3 x 3

16. **The value of** $\begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 0 \\ 2 & -3 & 5 \end{vmatrix}$ **is**

(A) 0 (B) 2 (C) -2 (D) 3

17. **If** $A=\begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$ **then order of A^t is**

(A) 3 x 1 (B) 1 x 3 (C) 3 x 3 (D) 1 x 1

18. **If** $A=\begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$ **then** $|A|$ **equals .**

(A) 1 (B)0 (C) – 1 (D) 12

19. **If the order of a matrix A is m x n and order of matrix B is n x p then the order AB is**

(A) p x n (B) m x p (C) p x m (D) n x p

20. **If** $A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ **and B =** $\begin{bmatrix} ka_{11} & ka_{12} \\ ka_{21} & ka_{22} \end{bmatrix}$**, then** $|B|$ **is equal to**

(A) $k|A|$ (B) $k^2|A|$ (C) $\frac{1}{k}|A|$ (D) $\frac{1}{k^2}|A|$

21. **If A and B are non – singular matrices then** $(AB)^{-1}$ **equals**

(A) $A^{-1}B^{-1}$ (B) $\frac{1}{AB}$ (C) $B^{-1}A^{-1}$ (D) $(BA)^{-1}$

22. **The trivial solution of the homogenous linear equations is**

(A) (0,0,0) (B)(1, 0, 0) (C) (0,1,0) (D) (0,0,1)

23. **If A is a matrix of order m x n, then the number of elements in each row of A is**

(A) m (B)n (C) m + n (D) m – n

24. **If A is a square matrix of order 3 x 3 then** $|kA|$ **equals**

(A) $k|A|$ (B) $k^2|A|$ (C) $k^3|A|$ (D) $k^4|A|$

25. **For matrix equation** $\begin{bmatrix} 3 & 1 \\ -3 & 3y-4 \end{bmatrix}=\begin{bmatrix} 3 & 1 \\ -3 & 2 \end{bmatrix}$ **then value of y =**

(A)1 (B)2 (C) 3 (D) 4

26. **If A is a matrix of order 3 x 2 then order of A^tA =**

(A) 3 x 3 (B) 2 x 3 (C) 2 x 2 (D) 3 x 2

27. **If each element of a matrix is zero then it is called.**

(A) Hermitian (B) Symmetric (C) Null matrix (D) Skew symmetric

28. $(AB)^t =$

(A) A^t (B) B^t (C) B^tA^t (D) A^tB^t

29. **If order of a matrix A is 2 x 3 and of matrix B is 3 x 3 then order of AB is**

(A) 3 x 2 (B) 2 x 2 (C) 3 x 3 (D) 2 x 3

30. **If** $\begin{bmatrix} \lambda & 1 \\ -2 & 1 \end{bmatrix}$ **is singular, then** $\lambda =$

(A) 1 (B) 2 (C) -2 (D) -1

31. **A square matrix A is symmetric if A^t =.**

(A) -A (B) A (C) A^t (D) A^2

32. **The value of determinant** $\begin{vmatrix} 1 & 12 & 25 \\ 0 & 3 & 15 \\ 0 & 0 & 8 \end{vmatrix}$ **is.**

(A) 0 (B) 1 (C) 8 (D) 24

33. **If AB exists, then $(AB)^{-1}$ is:**

(A) $A^{-1} B^{-1}$ (B) $B^{-1} A^{-1}$ (C) AB (D) $A^{-1}B$

34. **The order of** $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ *is* :

(A) 0 (B) 3 x 1 (C) 2 x 2 (D) 1 x 2

35. **The matrix** $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 6 \end{bmatrix}$ **is:**

(A) diagonal (B) scalar (C) triangular (D) singular

36. **If matrix** $\begin{bmatrix} 2 & 3 \\ 4 & x \end{bmatrix}$ **is singular, then x is:**

(A) 3 (B) 4 (C) 2 (D) 6

37. **If A is non-singular square matrix, Then AA^{-1} equals:**

(A) A (B) A^{-1} (C) 0 (D) I

38. **If two rows of any square matrix are identical, then the value of determinant is:**

(A) 1 (B) -1 (C) 0 (D) 2

39. **If A is a square matrix of order 2 then** $|kA|$ **equals**

(A) $k|A|$ (B) $k^2|A|$ (C) $\frac{1}{k}|A|$ (D) $2k|A|$

40. **If** $A = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}$ **then order of A^t is**

(A) 3 x 2 (B)2 x 3 (C) 2 x 2 (D) 3 x 3

41. **Inverse of a square matrix A does not exist if A is:**

(A) Singular (B) Non-Singular (C) Unit (D) Diagonal

42. **If** $\begin{vmatrix} -1 & 3 \\ x & 1 \end{vmatrix} = 0$ **then x = :**

(A) 3 (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

43. **A square matrix A is skew Hermitian if** $(\overline{A})^t =$

(A) A (B) $\overline{A}$ (C) A^t (D) – A

44. **The cofactor of an element a_{ij} denoted by A_{ij} is**

(A) $(-1)^{i-j} M_{ij}$ (B) $(-1)^{i+j} M_{ij}$ (C) $(-1)^{i-j} A_{ij}$ (D) $(-1)^{ij} M_{ij}$

45. $\left(A^{-1}\right)^t$ **is equal to**

(A) A (B) $\left(A^{-1}\right)A^t$ (C) $\left(A^t\right)^{-1}$ (D) $-A^t$

46. **The rank of the matrix** $\begin{bmatrix} 1 & 0 & 3 \end{bmatrix}$ **is**

(A) 0 (B) **1** (C) **2** (D) **3**

47. **The (0, 0) solution of the system** $a_1x+b_1y=0$ **and** $a_2x+b_2y=0$ **is called**

(A) Consistent (B) Inconsistent (C) Non-trivial (D) Trivial

48. **If** $\begin{vmatrix} k & 4 \\ 4 & k \end{vmatrix} = 0$**, then** $k =$ _____

(A) 16 (B) 0 (C) ± 4 (D) 8

49. **If A is a matrix of order** 3×4 **then order of** AA^t **is:**

(A) 4×3 (B) 3×4 (C) 4×4 (D) 3×3

50. **If A and B are matrices then** $(AB)^t$ **is equal to:**

(A) B^tA^t (B) A^tB^t (C) AB (D) BA

51. **The inverse of a square matrix exists if A is**

(A) Singular (B) Non-singular (C) Symmetric (D) Rectangular

52. **If matrix** $\begin{bmatrix} x & 4 \\ 2 & 8 \end{bmatrix}$ **is singular then** $x =$:

(A) 0 (B) -1 (C) 2 (D) 1

53. **A matrix A is said to be symmetric if:**

(A) $A^t = -A$ (B) $A^t = A$ (C) $\left(\overline{A}\right)^t = A$ (D) $\left(\overline{A}\right)^t = -A$

54. **A matrix A is skew symmetric if** $A^t =$:

(A) $-A$ (B) A (C) $\overline{A}$ (D) A^t

55. **If order of a matrix A is** $m \times n$**, then order of** A^t **is:**

(A) $m \times n$ (B) $m \times m$ (C) $n \times m$ (D) $n \times n$

56. **If** $\begin{bmatrix} x-3 & 1 \\ -5 & -4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -5 & -4 \end{bmatrix}$**, then** $x =$ **:**

(A) 5 (B) -5 (C) -1 (D) 1

57. **If A=** $\begin{bmatrix} 1 & 2 & -2 \\ 0 & 0 & 5 \\ 6 & 7 & 3 \end{bmatrix}$**, then** $A_{33} =$:

(A) -1 (B) 1 (C) 3 (D) 0

58. **A square matrix** $A = [a_{ij}]$ **in which** $a_{ij} = 0$ **for all** $i > j$ **is called:**

(A) Upper triangular (B) Lower triangular (C) Symmetric (D) Skew Symmetric

59. **If** $A = [a_{ij}]_{2\times 2}$**, then** $|KA| =$:

(A) $|A|$ (B) $k^2|A|$ (C) $k|A|$ (D) $k|A|^2$

Multiple Choice Questions

1. For S = sum of roots & P = product of roots the quadratic equation is.
(A) $x^2 + Sx + P = 0$ (B) $x^2 + Sx - P = 0$ (C) $x^2 - Sx + P = 0$ (D) $x^2 - Sx - P = 0$

2. The sum of the all three cube roots of unity is
(A) 1 (B) -1 (C) 3 (D) 0

3. The polynomial ax^3+bx^2+8 has degree
(A)8 (B)3 (C)a+b (D)None

4. If w is a cube root of unity, the other root is.
(A)1/w (B)w (C)1 (D) None of these

5. One root of the equation $x^2+1=0$ is *i* & other root is.
(A)-1 (b)-i (C) i^2 (D) None of these

6. If x^3+3x^2-6x+2 is divided by x+2 then remainder is.
(A) -18 (B) 9 (C) -9 (D) 18

7. Two square roots of unity are
(A)1, -1 (B) -1 (C)w (D)-i,i^2

8. Roots of a quadratic equation are imaginary if b^2-4ac is
(A) < 0 (B) > 0 (C) = 0 (D) = 1

9. A quadratic equation has degree
(A)One (B) Two (C) Three (D) Four

10. If $f(x) = x^3 +4x^2-2x+5$ is divided by x-1 then remainder is:
(A) 8 (B) -8 (C) 10 (D) None of these

11. A reciprocal equation remains unchanged when variable x is replaced by
(A) 1/x (B) -1/x (C) $1/x^2$ (D) -x

12. The roots of the equation $ax^2+bx+c=0$ are rational if.
(A) $b^2-4ac < 0$ (B) $b^2-4ac > 0$ (C)$b^2-4ac = 0$ (D) $b^2-4ac>0$ & perfect square

13. The discriminant of quadratic equation $ax^2+bx+c=0$ is
(A) $4ac-b^2$ (B)$b^2 + 4ac$ (C) a^2-4ac (D) b^2-4ac

14. The product of the roots of equation $5x^2-x+2=0$ is
(A)5/2 (B) -5/2 (C)2/5 (D)2

15. If a polynomial $f(x)$ of degree $n\geq1$ is divided by (x-a) then the remainder R is
(A) $f(-a)$ (B) $f(a)$ (C) a $f(a)$ (D) $f(0)$

16. One root of the equation $x^2-3x+a=0$ is 2, then a is
(A) -2 (B) 2 (C) 3 (D) -3

17. The solution of the equation $ax^2+bx+c=0$ is
(A) $x = \frac{b \pm \sqrt{b^2+4ac}}{2a}$ (B) $x= \frac{-b \pm \sqrt{b^2+4ac}}{2a}$ (C) $x=\frac{-b \pm \sqrt{b^2-4ac}}{2a}$ (D) $x= \frac{b \pm \sqrt{b^2+4ac}}{2a}$

18. The roots of the equation $25x^2-30x+9 = 0$ are
(A) Imaginary (B) rational & unequal (C) rational & equal (D) irrational & equal

19. For what value of k will equation $x^2-kx+4=0$ have sum of roots equal to product of roots.
(A) 3 (B)-2 (C) -4 (D) 4

20. For equal roots of $ax^2+bx+c = 0$ (b^2-4ac) will be.
(A) Negative (B) 0 (C) 1 (D) 2

21. Sum of four fourth roots of unity is.
(A) 1 (B)-1 (C) 0 (D) –i

22. Equation having same solutions are called:
(A) Exponential equations (B)radical equations (C) simultaneous equations(D) reciprocal equations

23. If $4^x = 1/2$ then x equals.
(A)1/2 (B) -1/2 (C) 2 (D) -2

24. If polynomial x^2+3x+7 is divided by x+1 then remainder is:
(A) -5 (B) 5 (C)11 (D)-11

25. If $4^x = 2$ then x equals
(A)1/2 (B) -1/2 (C) 2 (D) -2

26. If α, β are the roots $ax^2+bx+c=0$ then αβ is equal to.
(A)-a/b (B) b/a (C) c/a (D) –b/a

27. The quadratic equation with roots $2-\sqrt{3}$ & $2+\sqrt{3}$ is
(A)$x^2+4x+1 = 0$ (B)$x^2-4x+1=0$ (C)$x^2-6x+6=0$ (D)$x^2-6x-6=0$

28. The sum & product of roots of equation $4x^2+7x+3= 0$ are
(A)3/4, -4/7 (B)-7/3, -3/4 (C)-7/4 , 3/4 (D)-3/7,-4/3

29. Roots of $x^2-x-2=0$ are:
(A)2,-1 (B) -2,1 (C)-2,-1 (D)2,1

30. Product of all three cube roots of unity is:
(A)1 (B)0 (C)-1 (D)w

31. The number of roots of the equation $8x^6-19x^3+27=0$ is
(A)2 (B) 3 (C)4 (D)6

32. The roots of the equation $ax^2+bx+c = 0$ are irrational if b^2-4ac is:
(A) Negative (B) zero
(C) positive & perfect square (D) Positive & not a perfect square

33. Let $f(x)=x^3+4x^2-2x+5$ is divided by x-1 then remainder equals.
(A)0 (B) 6 (C) 7 (D)8

34. The product of all fourth roots of unity equals.
(A)1 (B) 0 (C) -1 (D)2

35. If α & β are the roots of $3x^2-2x+4 = 0$ then the equation whose roots are 2α, 2β are
(A)$2x^2+6x+8=0$ (B)$4x^2-2x+3=0$ (C)$3x^2-4x+16=0$ (D)$3x^2+16x-4=0$

36. The roots of the equation $x^2+px+q=0$ are additive inverse of one another then.
(A)p=1 (B)q=1 (C)q=0 (D) p=0

37. If w is the imaginary cube root of unity then w^2 =
(A)1 (B) -1 (C)w^{-1} (D)w^{-2}

38. If polynomial x^2-2x+2 is divided by x-1, then remainder is:
(A)1 (B) -1 (C)0 (D)2

39. $1+w+w^2$=
(A)1 (B) -1 (C)w^{-1} (D) 0

40. The discriminant for equal roots is:
(A) > 0 (B) < 0 (C) = 0 (D) perfect square

41. Roots of equation $x^2-4x+8=0$ are
(A) Imaginary (B)Real (C)Rational (D) Equal

42. When $3x^4+4x^3+x-5$ is divided by $x+1$ then remainder is.
(A) 7 (B) 6 (C) -6 (D) -7

43. In a quadratic equation $ax^2+bx+c=0$ if $b^2-4ac<0$, then roots are:
(A) Real (B)Equal (C) Rational (D) Complex/imaginary

44. The polynomial $3x^2 + 2x + 1$ has degree:
(A) 0 (B) 3 (C) 2 (D) 4

45. **If $4^x = \frac{1}{2}$ then x = :**

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 2 (D) -2

46. **$1 + \omega^4$ = :**

(A) ω (B) $-\omega$ (C) ω^2 (D) $-\omega^2$

47. **Product of roots of $ax^2 + bx + c = 0$ is:**

(A) $-\frac{c}{a}$ (B) $\frac{c}{a}$ (C) $-\frac{a}{c}$ (D) $-\frac{b}{a}$

48. **The sum of the all four fourth roots of unity is:**

(A) 0 (B) 1 (C) -1 (D) i

49. **The Product of all three cube roots of unity is:**

(A)0 (B)1 (C)-1 (D)w

50. **If $b^2-4ac > 0$ and perfect square then roots are :**

(A)Rational (B)Irrational (C)Equal (D)Complex

51. **The sum of the all three cube roots of unity is**

(A) 1 (B) -1 (C) 3 (D) 0

52. **The roots of $x^2 + x - 6 = 0$ are**

(A) Real (B) Equal (C) Complex (D) Trivial

53. **The sum of roots of $ax^2 + bx + c = 0$, $a \neq 0$ is:**

(A) $\frac{b}{a}$ (B) $\frac{c}{a}$ (C) $-\frac{b}{a}$ (D) $-\frac{c}{a}$

54. **The sum of the all four fourth roots of unity is:**

(A) 0 (B) 1 (C) -1 (D) i

55. **If ω is complex cube root of unity then conjugate of ω is :**

(A) ω^2 (B) $-\omega^2$ (C) $-\omega$ (D) i

56. **If ω is cube root of unity then ω = :**

(A) ω^2 (B) $1-\omega^2$ (C) ω^{-2} (D) $1+\omega^2$

57. **If one solution of the equation $x^2 + ax + 2 = 0$ is $x = 1$, then $a =$:**

(A)**-7** (B) 7 (C)**-3** (D) 0

58. **If ω is the imaginary cube root of unity then ω^2 = :**

(A) **1** (B) ω^{-2} (C) $-\omega-1$ (D) $1+\omega$

59. **If α, β are the roots of equation of $ax^2 - bx + c = 0$, then $\alpha + \beta =$:**

(A) $\frac{b}{a}$ (B) $\frac{a}{b}$ (C) $\frac{c}{a}$ (D) $-\frac{b}{a}$

60. **The roots of equation $x^2 - 5x + 6 = 0$ are:**

(A) $2, -3$ (B) $-2, -3$ (C) $2, 3$ (D) $-2, 3$

61. **A quadratic equation has degree:**

(A) 0 (B) 1 (C) 2 (D) 3

62. **Product of all fourth roots of unity is :**

(A) -1 (B) 0 (C) 1 (D) i

63. **Roots of equation $x^2 - 7x + 10 = 0$ are:**

(A) $(2, -5)$ (B) $(-2, 5)$ (C) $(2, 5)$ (D) $(-2, -5)$

64. **If ω be cube root of unity, then ω^2 = :**

(A) $\frac{-1-\sqrt{3}i}{2}$ (B) $\frac{1-\sqrt{3}i}{2}$ (C) 1 (D) $\frac{1+\sqrt{3}i}{2}$

65. **The sum of four 4th roots of 16 is:**

(A) 0 (B) 2 (C) 4 (D) 16

66. **If $x - a$ is a factor of polynomial $f(x)$, then $f(a)$ is:**
(A) $= 0$ (B) < 0 (C) > 0 (D) $\neq 0$

67. **The sum of the four fourth roots of 81 is:**
(A) 0 (B) 81 (C) -81 (D) 3

68. **If $b^2 - 4ac > 0$ but not a perfect square, then roots are:**
(A) Equal (B) Complex (C) Rational (D) Irrational

CHAPTER-5 PARTIAL FRACTION

Multiple Choice Questions

1. **The rational fraction $\frac{P(x)}{Q(x)}$ is a proper fraction if:**
(A) Degree of P(x) = Degree of Q(x) (B) Degree of P(x) < degree of Q(x)
(C) Degree of Q(x) < degree of P(x) (D) None of these

2. **The identity $(x+3)(x+4)=x^2+7x+12$ is true for:**
(A) Two values of x (B) One value of x (C) All values of x (D) None of these

3. **The partial fraction of $\frac{x+5}{(x-1)(x^2+1)}$ are the form of:**
(A) $\frac{A}{x-1}+\frac{B}{x^2+1}$ (B) $\frac{A}{x-1}+\frac{Bx+C}{x^2+1}$ (C) $\frac{Ax}{x-1}+\frac{B}{x^2+1}$ (D) None of these

4. **The function of the form $f(x) = \frac{P(x)}{Q(x)}$, $Q(x)\neq 0$ where P(x) & Q(x) are polynomials is called:**
(A) Identity (B) Equation (C) Fraction (D) Algebraic relation

5. **Partial fraction of $\frac{1}{x^2-1}$ is equivalent to:**
(A) $\frac{Ax+B}{x^2-1}$ (B) $\frac{A}{x+1}+\frac{B}{x-1}$ (C) $\frac{A}{x+1}$ (D) $\frac{B}{x-1}$

6. **The fraction $\frac{x^2+7x+3}{x+1}$ is:**
(A) Improper (B) Proper (C) Equivalent (D) None of these

7. **Partial fraction of $\frac{1}{x^3+1}$ will be of the form**
(A) $\frac{A}{x-1}+\frac{B}{x^2-x+1}$ (B) $\frac{A}{x+1}+\frac{B}{x^2+1}$ (C) $\frac{A}{x+1}+\frac{Bx+C}{x^2-x+1}$ (D) $\frac{A}{x-1}+\frac{B}{x^2+1}$

8. **Partial fraction of $\frac{7x^2+25}{(x+3)(x+4)}$ will be of the form:**
(A) $\frac{A}{x+3}+\frac{B}{x+4}$ (B) $1+\frac{A}{x+3}+\frac{B}{x+4}$ (C) $7+\frac{A}{x+3}+\frac{B}{x+4}$ (D) None of these

9. **The function $\frac{2x^2+5}{x+3}$ *is***
(A) Proper (B) Improper (C) Irrational fraction (D) None of these

10. **The fraction in which the degree of numerator is greater than the equal to the degree of denominator is called:**
(A) a proper fraction (B) an improper fraction (C) an equation (D) an algebraic relation

11. $\frac{x^3+1}{(x-1)(x+2)}$ *is*
(A) Proper fraction (B) improper fraction (C) Identity (D) None of these

12. Partial fractions of $\frac{1}{(x+1)(x^2-1)}$ will be of the form.

(A) $\frac{A}{x+1}+\frac{Bx+C}{x^2-1}$ (B) $\frac{A}{x+1}+\frac{B}{x-1}+\frac{C}{(x+1)^2}$

(C) $\frac{A}{x+1}+\frac{B}{(x-1)^2}$ (D) $\frac{A}{(x-1)}+\frac{Bx+c}{(x+1)^2}$

13. Partial fractions of $\frac{x^2+1}{(x+1)(x-1)}$ are of the form.

(A) $\frac{A}{x+1}+\frac{B}{x-1}$ (B) $1+\frac{A}{x+1}+\frac{B}{x-1}$ (C) $1+\frac{A}{x+1}+\frac{Bx+C}{x-1}$ (D) $\frac{Ax+B}{x+1}+\frac{C}{x-1}$

14. Types of rational fractions are:

(A) Three (B) Two (C) Four (D) Infinite

15. $\frac{1}{(x^2+1)(x-1)}=$?

(A) $\frac{A}{x^2+1}+\frac{Bx+C}{x+1}$ (B) $\frac{A}{x^2+1}+\frac{B}{x+1}$ (C) $\frac{Ax+B}{x^2+1}$ (D) $\frac{Ax+B}{x^2+1}+\frac{C}{x-1}$

16. $\frac{x^4}{1-x^4}$ is :

(A) Proper fraction (B) improper fraction (C) Decimal (D) Equation

17. Partial fractions of $\frac{1}{x^3-1}$ are of the form

(A) $\frac{A}{x-1}+\frac{Bx+C}{x^2+x+1}$ (B) $\frac{A}{x+1}+\frac{Bx+c}{x-1}+\frac{C}{(x+1)^2}$ (C) $\frac{A}{x-1}+\frac{Bx+C}{(x-1)^2}$ (D) $\frac{A}{x+1}+\frac{Bx+C}{x^2-x+1}$

18. Partial fraction of $\frac{1}{(x-1)^2(x+1)}$ is of the form.

(A) $\frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{C}{x+1}$ (B) $\frac{A}{x+1}+\frac{B}{x-1}+\frac{C}{(x+1)^2}$

(C) $\frac{A}{x-1}+\frac{Bx+C}{(x-1)^2}$ (D) $\frac{A}{x+1}+\frac{Bx+C}{x-1}+\frac{Dx+E}{(x-1)^2}$

19. Partial fraction of $\frac{(x-a)(x-b)}{(x-c)(x-d)}$ will be of the form:

(A) $\frac{A}{x-c}+\frac{B}{x-d}$ (B) $1+\frac{A}{x-a}+\frac{B}{x-b}$ (C) $\frac{A}{x-a}+\frac{B}{x-b}$ (D) $1+\frac{B}{x-c}+\frac{A}{x-d}$

20. Partial fraction of $\frac{x}{(x-1)(x+2)}$ will be of the form:

(A) $\frac{A}{x-1}+\frac{B}{x+2}$ (B) $\frac{1}{x-1}$ (C) $\frac{1}{x+2}$ (D) $1+\frac{A}{x-1}+\frac{B}{x+2}$

21. Partial fraction of $\frac{9x-7}{(x^2+1)(x+3)}$ are of the form:

(A) $\frac{Ax+B}{x^2+1}+\frac{C}{x+3}$ (B) $\frac{A}{x^2+1}+\frac{B}{x+3}$ (C) $\frac{A}{x^2+1}$ (D) $\frac{B}{x+3}$

22. Partial fractions of $\frac{1}{(x+1)(x^2-1)}$ will be of the form.

(A) $\frac{A}{x+1}+\frac{Bx+c}{x^2-1}$ (B) $\frac{A}{x+1}+\frac{B}{x-1}+\frac{C}{(x+1)^2}$

(C) $\frac{A}{x+1}+\frac{B}{(x-1)^2}+\frac{C}{x-1}$ (D) $\frac{A}{x-1}+\frac{Bx+C}{(x+1)^2}$

23. The function of the form $\frac{p(x)}{q(x)}$, q(x) ≠ 0 is called:

(A) identity (B) equation (C) fraction (D) formula

24. Partial fraction of $\frac{2}{x^2-1}$ are:

(A) $\frac{1}{x-1}+\frac{1}{x+1}$ (B) $\frac{1}{x-1}-\frac{1}{x+1}$ (C) $\frac{1}{2(x+1)}-\frac{1}{2(x-1)}$ (D) $\frac{1}{2(x-1)}-\frac{1}{2(x+1)}$

25. Partial fraction of $\frac{7x+25}{(x+3)(x+4)}$ will be of the form:

(A) $\frac{A}{x+3}+\frac{Bx}{x+4}$ (B) $\frac{A}{x+3}+\frac{B}{x+4}$ (C) $\frac{Ax+B}{x+3}+\frac{C}{x+4}$ (D) $\frac{Ax+B}{(x+3)(x+4)}$

26. Partial fractions of $\frac{1}{x(x+1)}$ are of the form

(A) $\frac{1}{x-1}+\frac{1}{x+1}$ (B) $\frac{1}{x-1}-\frac{1}{x+1}$ (C) $\frac{1}{x}+\frac{1}{x+1}$ (D) $\frac{1}{x}-\frac{1}{x+1}$

27. Types of rational fractions are:

(A) 3 (B) 2 (C) 4 (D) 1

28. An improper fraction can be reduced to proper fraction by :

(A) Addition (B) Subtraction (C) Multiplication (D) Division

29. $\frac{x^2}{(x^2-1)(x+1)}=\frac{A}{(x-1)}+\frac{B}{x+1}+:$

(A) $\frac{C}{(x+1)}$ (B) $\frac{C}{(x-1)}$ (C) $\frac{C}{(x+1)^2}$ (D) $\frac{x+1}{x^2+1}$

30. $x^2-x-2=(x+1)(x-2)$ **is :**

(A) Quadratic equation (B) Conditional equation (C) Linear equation (D) Identity

31. The fraction $\frac{3x^2+5}{x+1}$ is:

(A) Proper fraction (B) Polynomial (C) Partial fraction (D) improper fraction

32. From the identity $5x+4=A(x-1)+B(x+2)$, then value of $B=$:

(A) -3 (B) 3 (C) -2 (D) 2